

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0129

Parkin 138 kV
April 26, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0129. The Interconnection Customer proposes a 90.0 MW wind facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on May 11, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 109 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,935,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Parkin – Candler Mountain 138.0 kV, the Parkin – Opossum Crk 138.0 kV loads to 171.5% of its 209 MVA rating. Project DFAX: 100.0%.

For the loss of Parkin – Candler Mountain 138.0 kV, the Central Tie – Opossum Crk 138.0 kV loads to 161.5% of its 216 MVA rating. Project DFAX: 100.0%.

The Central Tie – Wythe 138.0 kV L is overloaded to 152.7% of its 222 MVA emergency rating for the loss of the Parkin – Candler Mtn 138.0 kV line. The project contributes 100.0% DFAX (90.0 MW) to this overload.

South Lynchburg – Brookville 138.0 kV: 128.2% loading (256 MVA) following the outage of the Parkin – Opossum Creek 138.0 kV L; DFAX = 100.0%.

The Brookville – Amargo 138.0 kV is overloaded to 127.5% of its 231 MVA emergency rating for the loss of the Parkin – Opossum Crk 138.0 kV. The project contributes 100.0% DFAX (90.0 MW) to this overload.

For the loss of Opossum Creek – Parkin 138.0 kV, the Candler Mtn – S Lynchburg 138.0 kV loads to 126.4% of its 267 MVA rating. Project DFAX: 100.0%.

For the loss of Parkin – Opossum Crk 138.0 kV, the Parkin – Candler Mtn 138.0 kV loads to 123.3% of its 291 MVA rating. Project DFAX: 100.0%.

The Skimmer – Pentland 138.0 kV L is overloaded to 123.0% of its 209 MVA emergency rating for the loss of the Clover – Bear 345.0 kV. The project contributes 5.3% DFAX (4.8 MW) to this overload.

Snowden – Skimmer 138.0 kV: 122.8% loading (210 MVA) following the outage of the Bear – Clover 345.0 kV; DFAX = 5.3%.

For the loss of Parkin – Opossum Creek 138.0 kV, the Amargo – Boonsboro 138.0 kV loads to 113.6% of its 241 MVA rating. Project DFAX: 100.0%.

For the loss of Leavitt – Birchwood 345.0 kV, the Bear Island – Dunn Siding 345.0 kV loads to 108.5% of its 1,303 MVA rating. Project DFAX: 14.1%.

For the loss of Bear – Clover 345.0 kV, the Shenandoah – Aibonito Pds 138.0 kV loads to 105.3% of its 238 MVA rating. Project DFAX: 7.5%.

Oak Crk – Carmel 146Th St 138.0 kV: 105.0% loading (156 MVA) following the outage of the Bear – Clover 345.0 kV line; DFAX = 6.5%.

Carmel 146Th St – Ceiba Baja 7012 138.0 kV: 102.0% loading (164 MVA) following the outage of the Clover – Bear 345.0 kV; DFAX = 6.5%.

S Crk – Wythe Substation 138.0 kV: 100.6% loading (286 MVA) following the outage of the Candler's Mtn – Parkin 138.0 kV L; DFAX = 46.1%.

Walworth – Shenandoah 138.0 kV: 100.1% loading (231 MVA) following the outage of the Bear – Clover 345.0 kV; DFAX = 7.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Parkin – Opossum Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$2,587,000	100% (\$2,587,000)
Central Tie – Opossum Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$1,888,000	100% (\$1,888,000)
Central Tie – Wythe Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$2,861,000	100% (\$2,861,000)
South Lynchburg – Brookville 138.0 kV	Rebuild line with higher-capacity conductor	\$3,079,000	100% (\$3,079,000)
Brookville – Amargo 138.0 kV	Rebuild line with higher-capacity conductor	\$1,016,000	100% (\$1,016,000)
Candler's Mtn – S Lynchburg 138.0 kV	Rebuild line with higher-capacity conductor	\$2,990,000	100% (\$2,990,000)
Parkin – Candler's Mtn 138.0 kV	Rebuild line with higher-capacity conductor	\$1,583,000	100% (\$1,583,000)
Skimmer – Pentland 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,531,000	100% (\$4,531,000)
Snowden – Skimmer 138.0 kV	Rebuild line with higher-capacity conductor	\$3,759,000	100% (\$3,759,000)
Boonsboro – Amargo 138.0 kV	Rebuild line with higher-capacity conductor	\$3,796,000	100% (\$3,796,000)
Bear Island Substation – Dunn Siding 345.0 kV	Reconductor line	\$36,913,000	2% (\$674,000)
Shenandoah – Aibonito Pds 138.0 kV	Reconductor line	\$10,757,000	100% (\$10,757,000)
Oak Crk – Carmel 146Th St 138.0 kV	Reconductor line	\$2,466,000	100% (\$2,466,000)
Ceiba Baja 7012 – Carmel 14Th St 138.0 kV	Reconductor line	\$3,171,000	100% (\$3,171,000)
Wythe – S Crk 138.0 kV	Reconductor line	\$3,251,000	33% (\$1,075,000)
Walworth – Shenandoah 138.0 kV	Reconductor line	\$10,702,000	100% (\$10,702,000)
	Total allocated to project		\$56,935,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.